

EVOLVING SAFETY LANDSCAPE OF MULTI-MODAL LARGE LANGUAGE MODELS: A SURVEY OF EMERGING THREATS AND SAFEGUARDS

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ABSTRACT

Multi-modal large language models (MLLMs) integrate heterogeneous modalities through modality alignment and fusion, enabling stronger understanding and reasoning. However, this architectural shift reshapes the safety landscape of machine learning. Increased model complexity and cross-modal interactions give rise to novel threats, including compromised modality integration, modality misalignment, and fused safety risks, reflecting shifts in threat modeling beyond uni-modal assumptions. These shifts, in turn, impose new constraints on safety solutions not captured by existing frameworks rooted in uni-modal learning. Motivated by these challenges, this survey provides a systematic analysis of the evolving safety landscape of MLLMs. We first propose a multimodal grounded taxonomy of safety threats and analyze shifts in threat models, covering adversarial attacks, data poisoning, jailbreaks, and hallucinations. We then summarize updated safety assumptions and organize recent advances in MLLM safety strategies accordingly. Finally, we discuss open challenges and future directions to inform the development of more principled and scalable safety mechanisms for multimodal systems.

1 INTRODUCTION

Modern multi-modal learning employs large models, such as large language models (LLMs), to integrate diverse modalities (e.g., text, image, audio, and video) for enhanced understanding and decision-making (Li et al., 2022a; Alayrac et al., 2022; Liu et al., 2023b; Brown et al., 2020; DeepSeek-AI, 2025; team, 2024), enabling applications in healthcare, autonomous driving, and legal decisions. Its foundation lies in modality alignment, which maps heterogeneous features into a shared representation space, and modality fusion, which integrates aligned information for more comprehensive and accurate reasoning (Yin et al., 2024a; Wang et al., 2023b; Zhu et al., 2023b; Xu et al., 2023; Baltrušaitis et al., 2018).

With the shift toward multi-modal architectures, the safety landscape of machine learning is undergoing a significant transformation. The unique properties of multi-modal learning introduce new safety challenges: (1) additional modalities expand the attack surface by introducing their own vulnerabilities; (2) cross-modal interactions may exhibit semantic inconsistencies and can be exploited to induce adversarial misalignment; and (3) the fusion process can be exploited, where malicious signals that appear benign in isolation may trigger harmful behavior when combined. These emerging threats reflect a shift in threat modeling, where attackers require only partial system access and can exploit surfaces beyond the input space, leading to cross-modal vulnerabilities.

These shifts further introduce new constraints on safety solutions that go beyond classic uni-modal assumptions. For example, defenses can no longer presume which modality is compromised or the specific type of threat, and must instead align with core multi-modal objectives such as modality alignment and fusion, while also monitoring internal model stages. Addressing these constraints calls for safety mechanisms that are both effective and cost-efficient, including safety-aware fine-tuning that preserves inter-modal coherence, safety-integrated preference optimization, and training-free inference-time solutions.

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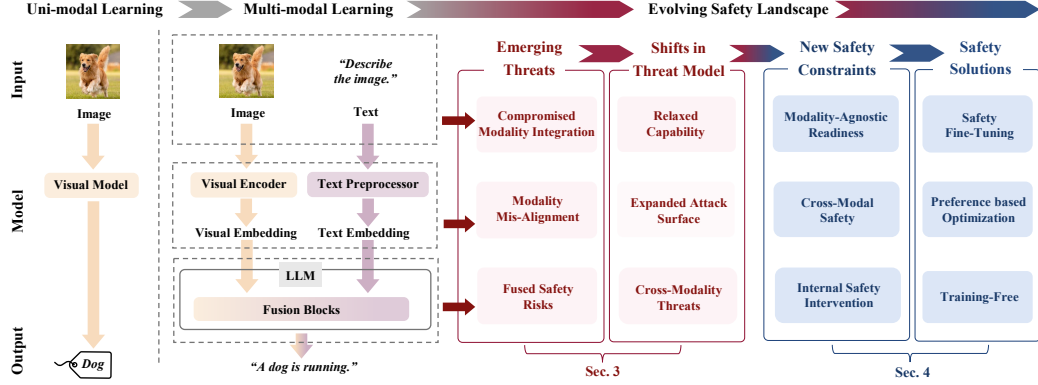


Figure 1: A multi-modal perspective on the evolving safety landscape, illustrated with Vision-Language LLM.

These changes motivate a comprehensive survey on the safety of multi-modal large language models (MLLMs). Recent surveys, summarized in Table 1, provide valuable overviews of attacks, defenses, and evaluation protocols (Liu et al., 2024e; Wang et al., 2024d; Fan et al., 2024; Wang et al., 2024e; Jin et al., 2024; Ye et al., 2025; Liu et al., 2025b), but largely adopt uni-modal taxonomies that organize studies by training/test threats or by separating vision-side and text-side attacks. Such perspectives may not fully capture risks unique to multi-modal systems, particularly those arising from cross-modal interactions. Moreover, prior surveys largely enumerate threats and defenses, with limited discussion of paradigm shifts in multi-modal safety, such as evolving threat models and safety constraints, offering limited guidance for future research on multi-modal safety.

Motivated by this gap, this survey provides a systematic analysis of the evolving safety landscape shaped by MLLMs (Figure 1), with the core contributions outlined as follows:

- We propose a new taxonomy of emerging safety threats in MLLMs arising from complex model architectures and cross-modal interactions, including compromised modality integration, cross-modal misalignment, and risks at the fusion stage. We further summarize the evolving threat modeling reflected by these threats.
- We summarize updated assumptions required for safety solutions in multi-modal settings and categorize recent advances in multi-modal safety within this framework, including supervised safety fine-tuning, preference-based optimization, and training-free methods.
- As safety research shifts from uni-modal to multi-modal systems, we highlight the need to address the increased structural complexity and interaction dynamics of MLLMs, and outline key directions for future research.

Although we discuss MLLM safety broadly, much of the literature focuses on vision-language models (VLMs), the most widely studied instantiation. Nevertheless, our key concepts generalize to MLLMs with additional modalities (e.g., audio, video, and sensory inputs). The remainder of the paper is organized as follows: Section 2 reviews uni-modal safety foundations; Section 3 presents MLLM characteristics, a taxonomy of emerging threats, and evolving threat models; Section 4 summarizes updated safety assumptions and recent safety solution advances; Section 5 discusses future directions; and Section 6 concludes.

2 EXISTING SAFETY LANDSCAPE

To provide necessary context, we review the well-established safety landscape shaped primarily by uni-modal learning. We follow standard attacker access definitions: white-box (full model knowledge), black-box (query-only access), and gray-box (partial knowledge, e.g., training data, architecture without parameters, or logits).

2.1 ADVERSARIAL ATTACKS

Adversarial attacks seek to induce incorrect outputs at inference time by adding imperceptible perturbations to inputs. Attackers manipulate test-time inputs without access to training data, operating in either black-box (Chen et al., 2017a) or white-box (Goodfellow et al., 2015; Madry et al., 2018a; Carlini & Wagner, 2017) settings. Such perturbations are typically generated via optimization-based methods that either target specific incorrect outputs or maximize prediction error, while satisfying norm-based constraints (e.g., l_∞ , l_2) to preserve perceptual imperceptibility, especially for images.

Survey	Venue	Threats				Solutions	MM Tax.	Safety Evo	Key Contributions
		A	P	J	H				
Liu et al. (2024e)	IJCAI	✓		✓		✓			Concise overview and basic categorization of VLM security
Fan et al. (2024)	SMC	✓	✓			✓			Overview of image-induced attacks and defenses
Wang et al. (2024e)	FCS		✓	✓	✓	✓			VLM security: attacks, defenses, and evaluation
Jin et al. (2024)	arXiv			✓		✓			Comprehensive survey of jailbreak in LLMs and MLLMs
Wang et al. (2024d)	EMNLP			✓		✓			Evolving Jailbreak landscape from LLMs to MLLMs
Ye et al. (2025)	arXiv	✓	✓	✓		✓			Lifecycle-based classification of VLM security across attacks, defenses, and evaluation
Liu et al. (2025b)	TNNLS	✓	✓	✓					Overview of VLM attacks and evaluation datasets/models
Ours	-	✓	✓	✓	✓	✓	✓	✓	MM-grounded taxonomy of MLLM safety and in-depth analysis of safety landscape shifts

Table 1: Recent surveys on MLLM safety. A: adversarial attacks; P: poisoning attacks; J: jailbreak attacks; H: hallucination; MM Tax.: MM-grounded taxonomy; Safety Evo: safety landscape evolution; VLM: vision-language model.

Defenses aim to enhance robustness against subtle perturbations, typically assuming full access to the victim model, sometimes with a small clean validation set. Adversarial training Madry et al. (2018b) improves robustness by iteratively generating adversarial examples and training the model to correctly classify them, while random smoothing Cohen et al. (2019); Li et al. (2019) injects Gaussian noise and averages predictions to smooth decision boundaries against small perturbations.

2.2 DATA POISONING

Data poisoning manipulates model behavior by injecting malicious training samples, assuming attacker access to the training set. Attacks range from label flipping Xiao et al. (2012); Zhang et al. (2021) that degrades overall performance to backdoor attacks that induce targeted misbehavior via trigger patterns while preserving clean accuracy. Triggers can take various forms, including subtle perturbations Nguyen & Tran (2021; 2020); Saha et al. (2020), imperceptible visual patterns in images Gu et al. (2019); Chen et al. (2017b), and specific tokens Li et al. (2021a), sentences Dai et al. (2019), or writing styles Qi et al. (2021) in text.

Poisoning defenses aim to reduce the impact of poisoned data while preserving model utility. Pre-training defenses detect and remove suspicious samples from the training set Tran et al. (2018); Paudice et al. (2018), training-time defenses leverage white-box access to select reliable samples or reduce poisoning effects during optimization Diakonikolas et al. (2019); Shen & Sanghavi (2019); Li et al. (2021c); Levine & Feizi (2021); Wang et al. (2022), and post-training defenses use white-box access and a small clean set to detect poisoned models Wang et al. (2019; 2023c), identify poisoned inputs Gao et al. (2019); Li et al. (2022b), or enforce benign behavior Liu et al. (2018); Li et al. (2021b); Zeng et al. (2022); Li et al. (2025a).

2.3 JAILBREAK

Jailbreak attacks bypass LLM safety by crafting prompts that elicit unauthorized rather than misleading outputs. White-box attacks optimize adversarial prefixes or suffixes Zou et al. (2023); Jones et al. (2023); Zhu et al. (2023c), Gray-box attacks exploit logit access to manipulate prompts and token selection Guo et al. (2024); Zhao et al. (2024b), or perform limited retraining via small-scale poisoning Qi et al. (2024b); Zhan et al. (2024); Yang et al. (2023). and black-box attacks leverage role-playing or contextual reasoning to bypass safety alignment Wang et al. (2023a); Wei et al. (2023); Li et al. (2023c). Other methods include LLM-generated adversarial prompts Deng et al. (2024); Liu et al. (2024d) and niche-language evasion Yuan et al. (2024).

Jailbreak defenses seek to maintain safety alignment. Black-box defenses filter or preprocess adversarial prompts to neutralize harmful intent Alon & Kamfonas (2023); Jain et al. (2023), while white-box defenses impose stronger guardrails via safety fine-tuning Bianchi et al. (2024); Deng et al. (2023), reinforcement learning from human feedback (RLHF) Ouyang et al. (2022), or model self-correction to reduce unsafe outputs Sun et al. (2023b).

2.4 HALLUCINATION

Hallucination refers to the generation of confident but factually incorrect or unsupported information Huang et al. (2025b), arising from limitations in training data, objectives, or decoding. It is mainly a reliability issue rather than a security issue involving active attackers. Common causes include noisy or biased data Bender et al. (2021); Sheng et al. (2020); Wallace et al. (2019), overfitting to spurious language patterns McKenna et al. (2023); Wang et al. (2021); Tu et al. (2020), and limited uncertainty modeling Farquhar et al. (2024).

To mitigate hallucination, some works improve data quality through filtering, deduplication, and high-quality structured or instructional data (Penedo et al., 2023; Lee et al., 2021; Guu et al., 2020;

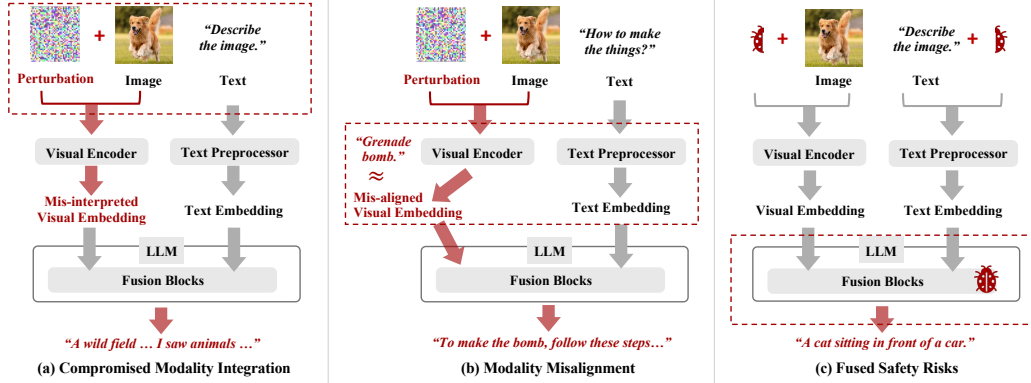


Figure 2: Illustration of (a) compromised visual integration (Gao et al., 2024); (b) targeted misalignment for jailbreaks (Shayegani et al., 2024); (c) fused backdoor attacks (Walmer et al., 2022).

Raffel et al., 2020). Other works adopt RLHF, contrastive learning, factual-consistency losses, or chain-of-thought supervision (Ziegler et al., 2019; Sun et al., 2023a; Wei et al., 2022). Additional works employ retrieval-augmented generation (RAG), prompt constraints, and post-hoc verification using confidence estimation or external tools (Lewis et al., 2020; Izacard & Grave, 2020; Zhou et al., 2022; Beurer-Kellner et al., 2023; Gao et al., 2023; Farquhar et al., 2024; Li et al., 2023d).

3 EMERGING MULTI-MODAL SAFETY THREATS

To integrate diverse data sources and enable cross-modal reasoning, MLLMs rely on two core mechanisms: modality alignment and modality fusion (Fig. 1). Modality alignment projects heterogeneous inputs into a shared embedding space using modality-specific encoders while preserving essential features, whereas modality fusion combines aligned representations (e.g., via cross-attention) to capture cross-modal semantics and support downstream reasoning tasks. However, the inherent complexity of these mechanisms and dynamic cross-modal interactions give rise to a new set of safety threats differ fundamentally from uni-modal settings. To characterize these emerging threats, we propose a new taxonomy of multi-modal safety risks (Fig. 2): Compromised Modality Integration, Modality Misalignment, and Fused Safety Risks.

3.1 COMPROMISED MODALITY INTEGRATION

When multiple modalities are integrated, threats may originate from **individual modalities**, where manipulation of one or more inputs propagates through the integration process and compromises overall model behavior. We refer to such threats as Compromised Modality Integration (Fig. 2 (a)).

Adversarial attacks in MLLMs primarily target the vision encoder by inserting visual perturbations that disrupt its representations. These errors then propagate through the alignment and fusion processes, compromising the behavior of the entire system. For example, Luo et al. (2024) and Schlarman & Hein (2023) craft visual perturbations that induce MLLMs to generate attacker-specified text. Several works Yin et al. (2023); Wang et al. (2024a;g) maximize the embedding distance between clean and perturbed images, distorting the model’s understanding. Gao et al. (2024) introduce perturbations that delay the end-of-sequence token, increasing uncertainty in output length. Bailey et al. (2024) show that adversarial images can bypass safety filters, leak context, and trigger misleading. Fu et al. (2023) demonstrate that subtle perturbations can covertly induce MLLM agents to perform malicious actions, such as leaking conversations. Cai et al. (2025a) boost transferability by applying visual transformations to ensure all perturbed variants fool the model. Zhang et al. (2025b) improve scalability, by using a self-supervised setup where a random image generates perturbations guided by the original. Zhang et al. (2024a) propose a benchmark of perturbed inputs that consistently trigger low-quality MLLM responses.

Liu & Zhang (2025) poison the visual encoder to map backdoored images to a target representation, inducing MLLM outputs aligned with the target content. MLLMs inherit jailbreak vulnerabilities from their LLM core, making them susceptible to harmful or unauthorized outputs Liu et al. (2024d); Yu et al. (2024b); Deng et al. (2024). Attackers can manipulate non-text modalities, such as image, to bypass safety mechanisms of LLM, triggering jailbreak responses with benign text and malicious visuals. For example, Qi et al. (2024a) optimize a visual adversarial example to universally jailbreak an aligned LLM. FigStep Gong et al. (2025) embeds rephrased jailbreak prompts within images to

Category	Sub-Category	Paper	Threat Model			Year & Venue
			Cap. 1	Cap. 2	Cap. 3	
Compromised Modality Integration	Adversarial Attacks	CroPALuo et al. (2024)	✓			ICLR'24
		Verbose ImagesGao et al. (2024)	✓			ICLR'24
		Image HijacksBailey et al. (2024)			✓	ICML'24
		AnyAttackZhang et al. (2025b)	✓			CVPR'25
	Poisoning	BadVisionLiu & Zhang (2025)	✓			CVPR'25
	Jailbreak	Adversarial Vision JailbreaksQi et al. (2024a)	✓		✓	AAAI'24
		Jailbreak In PiecesShayegani et al. (2024)	✓	✓	✓	ICLR'24
		FigStepGong et al. (2025)	✓		✓	AAAI'25
Hallucination	LLaVA-NextLiu et al. (2024c)	-	-	-	arXiv'24	
	Prismatic VLMsKaramcheti et al. (2024)	-	-	-	ICML'24	
Modality Mis-alignment	Untargeted Mis-alignment	Pandora's BoxLiu et al. (2024a)		✓		NeurIPS'24
		Break the Visual PerceptionWang et al. (2024g)	✓			ACM MM'24
		TMM AttackWang et al. (2024a)		✓		IEEE S & P'24
	Targeted Mis-alignment	VLM Adversarial RobustnessZhao et al. (2023a)		✓		NeurIPS'23
		Adversarial IllusionsBagdasaryan et al. (2024)		✓		USENIX Security'24
		ShadowcastXu et al. (2024a)		✓		NeurIPS'24
		BadCLIPBai et al. (2024)		✓		CVPR'24
		VLM Safety EvaluationTu et al. (2024)		✓		ECCV'24
		TrojVLMlyu et al. (2024)		✓		ECCV'24
		Revisit VLM BackdoorLiang et al. (2025)		✓		CVPR'25
		VLOODLYu et al. (2025)		✓		ICLR'25
		BadSemZhong et al. (2025)		✓		arXiv'25
Semantic Inconsistency	Minigt-4Zhu et al. (2023a)	-	-	-	arXiv'23	
	Blip-2Li et al. (2023a)	-	-	-	ICML'23	
	InstructBLIPDai et al. (2023)	-	-	-	arXiv'23	
	mPLUG-Owl2Ye et al. (2024)	-	-	-	CVPR'24	
Fused Safety Risks	Fusion-triggered threats	Dual-Key BackdoorWalmer et al. (2022)		✓		CVPR'22
		AnyDoor AttackLu et al. (2024)		✓	✓	arXiv'24
		ImgTrojanTao et al. (2025)	✓	✓	✓	NAACL'25
	Modality Conflicts	Curse of Multi-ModalitiesLeng et al. (2024)	-	-	-	arXiv'24

Table 2: Representative studies of multi-modal safety threats under the new taxonomy. Cap. 1: relaxed capability; Cap. 2: expanded attack surface; Cap. 3: cross-modality threats.

deceive the model. Shayegani et al. (2024) craft visual perturbations that bring visual signals close to jailbreak content in the embedding space.

During multi-modal integration, perceptual flaws in individual modalities can propagate and lead to system-wide hallucinations. For instance, low-resolution images may obscure small but critical objects, or speech recognition may omit key words. LLaVA-Next Liu et al. (2024c) shows that higher-resolution inputs reduce hallucinations, while Prismatic Karamcheti et al. (2024) leverages expert encoders to preserve fine-grained details. When such information is lost, the model tends to rely on pretraining priors to “fill in the gaps,” often resulting in hallucinated content.

3.2 MODALITY MISALIGNMENT

Adversaries can manipulate **cross-modal embeddings** to disrupt semantic or structural alignment between modalities, misleading the model during inference (Fig. 2(b)). Adversarial misalignment can be: (1) *Untargeted*, where the perturbed modality’s embedding diverges from clean modalities; or (2) *Targeted*, where embeddings are manipulated to mimic harmful representations. Misalignment may also arise naturally due to inherent inconsistencies across modalities.

For *untargeted* misalignment, existing works generally maximize cross-modal discrepancy/distance in the shared embedding space, thereby disrupting alignment between modalities. For example, Liu et al. (2024a) generate a universal adversarial patch by minimizing cosine similarity between visual and textual embeddings. Wang et al. (2024g) introduce visual perturbations that disrupt relations among visual tokens or weaken global visual-text semantic alignment. Wang et al. (2024a) focus on suppressing features that promote modality consistency while amplifying those that increase cross-modal discrepancy.

For *targeted* misalignment, Zhao et al. (2023a) align adversarial image embeddings with (1) the target text, (2) embeddings of corresponding target images, or (3) model outputs associated with the target text. Shayegani et al. (2024) optimize visual perturbations to mimic embeddings of harmful content (e.g., OCR-decoded jailbreak prompts), enabling cross-modal jailbreaks. Similarly, Qi et al. (2024a) craft adversarial visuals to maximize the likelihood of harmful text generation, breaking safety alignment. Bagdasaryan et al. (2024) and Tu et al. (2024) craft perturbations that minimize cosine similarity between the input and an unrelated adversarial target text. Furthermore, several works Lu et al. (2023); Han et al. (2023); Huang et al. (2025a) improve the transferability of embedding misalignment.

Some methods induce targeted misalignment via data poisoning. Lyu et al. (2024); Liang et al. (2025); Lyu et al. (2025) implant backdoors in MLLMs that trigger targeted harmful outputs when images contain backdoor patterns. Further, Xu et al. (2024a) poison the model to align embeddings

of original and destination concept images, causing it to generate text about the destination when shown the original. Bai et al. (2024) introduce sample-specific triggers and trigger-aware context prompts to shift visual embeddings toward a target class. Zhong et al. (2025) exploit semantic inconsistencies between paired modalities, e.g., mismatches in color or object descriptions, to trigger backdoor behaviors.

Inherent semantic inconsistencies across modalities can naturally cause misalignment and lead to hallucinations. Besides, modality alignment projects heterogeneous inputs into a shared embedding space, which is often anchored to text. Improper mappings may distort semantics and induce hallucinated outputs (Zhu et al., 2023a). For example, visual features from a medical scan may align with incorrect clinical concepts, causing the model to generate convincing but unfaithful outputs. Methods like BLIP-2 (Li et al., 2023a) and InstructBLIP (Dai et al., 2023) use query-based feature extraction to improve cross-modal faithfulness, but fail when queries miss critical visual details. Similarly, mPLUG-Owl2 (Ye et al., 2024) employs modality-adaptive attention, yet misalignment persists under domain shifts or out-of-distribution inputs.

3.3 FUSED SAFETY RISKS

Fused safety risks exploit the fusion mechanism, where adversarial signals are benign in isolation but **harmful when combined** during fusion, as shown in Fig. 2 (c). These threats manifest deeper in the model pipeline and are harder to detect. Beyond adversarial manipulation, naturally conflicting or imbalanced cross-modal evidence can also lead to unsafe behaviors during fusion.

Several studies illustrate fused safety risks. Walmer et al. (2022) embed backdoor triggers in both image and text, where each modality appears benign in isolation but induces malicious behavior when combined. Lu et al. (2024) show asymmetric modality roles, with images facilitating trigger injection and text activating malicious responses. Tao et al. (2025) associate harmful queries with specific clean images by replacing captions with jailbreak prompts, causing model generates harmful content when presented with both. Gong et al. (2025) embed jailbreak prompts in visual inputs and use benign-looking text to elicit harmful outputs. Song et al. (2025) approximate safety decision boundaries in multi-modal fusion spaces using logistic regression and manipulate fusion-layer features toward a target decision to induce policy-violating outputs.

Beyond attacks, modality fusion can also induce hallucinations when cross-modal evidence is imbalanced, allowing dominant linguistic priors to override weaker visual or auditory signals. This effect is empirically shown in Curse of Multi-Modalities (Leng et al., 2024), where MLLMs frequently default to unimodal dominance, leading to systematic hallucinations under fusion conflicts.

3.4 SHIFTS IN THREAT MODELS

Based on the above discussion of emerging threats, we summarize the key shifts in threat modeling introduced by multi-modal settings compared to uni-modal models. (1) **Relaxed Capability:** In multi-modal systems, adversaries no longer need full-system access. Due to the compositional design, compromising a subset of modalities can suffice. For example, adversarial images targeting only the visual encoder can trigger downstream misalignment during fusion, resulting in harmful or misleading outputs. (2) **Expanded Threat Surface:** Multi-modal systems introduce new threat surfaces beyond input-output mapping, including vulnerabilities in modality alignment and fusion. These internal stages create deeper, stealthier entry points for threats within the model pipeline. (3) **Cross-Modality Threats:** Multi-modal models enable threats that exploit interactions across modalities. For example, an adversarial image may trigger a jailbreak, or a malicious prompt may mislead the model’s interpretation of visual input.

4 CURRENT MULTI-MODAL SAFETY SOLUTIONS

In this section, we summarize new safety constraints in multi-modal systems driven by evolving threat models and review recent advances in safety fine-tuning, preference-based optimization, and training-free approaches.

4.1 NEW SAFETY CONSTRAINTS

(1) **Modality-Agnostic Readiness:** Multi-modal systems face dynamic threats across modalities, making fixed assumptions about threat types or entry modalities infeasible. For example, relying solely on text-based safety mechanisms is insufficient – adversarial visual inputs can bypass language-focused safeguards by embedding hidden jailbreak triggers, exposing critical blind spots in

Category	Sub-category	Paper	Safety Constraints			Year & Venue
			Cap. 1	Cap. 2	Cap. 3	
Safety Fine-Tuning	Modality-Wise Fine-Tuning	Robust CLIPSchlarmann et al. (2024)			✓	ICML'24
		Textual UnlearningChakraborty et al. (2024)			✓	EMNLP'24 Findings
	Cross-Modal Fine-Tuning	TGA-ZSR Yu et al. (2024a)		✓	✓	NeurIPS'24
		VLGuardZong et al. (2024)	✓	✓		ICML'24
		HallusionbenchGuan et al. (2024)	✓	✓		CVPR'24
		MmcLiu et al. (2024b)	✓	✓		NAACL'24
		Robust-VLGuardWang et al. (2025b)		✓		ICCV'25
		TGA Xu et al. (2025)		✓	✓	ICLR'25
		BYE Rong et al. (2025)	✓	✓		NeurIPS'25
		UNIGUARDOh et al. (2025)	✓			arXiv'25
		ProEATLu et al. (2025)	✓	✓		arXiv'25
Preference-Based Optimization	Human Feedback	SPA-VLZhang et al. (2025c)	✓	✓		CVPR'25
		AdPOLiu et al. (2025a)		✓	✓	arXiv'25
		GuardReasoner-VLLiu et al. (2025c)	✓	✓		NeurIPS'25
		Adversary-aware DPOWeng et al. (2025)	✓	✓	✓	EMNLP'25 Findings
	AI Feedback	SilkieLi et al. (2023b)	✓	✓		arXiv'23
Training-Free Solutions		MHALOCAi et al. (2025b)	✓	✓		ACL'25 Findings
	Input-level Safeguard	AdaShield Wang et al. (2024f)	✓	✓		ECCV'24
		BlueSuffix Zhao et al. (2025)	✓	✓		ICLR'25
		LLM-PipelineGuo et al. (2025)		✓		NeurIPS'25
	Internal-level Intervention	InferAlignerWang et al. (2024b)		✓	✓	EMNLP'24
		ASTRA Wang et al. (2025a)		✓	✓	CVPR'25
		ETADing et al. (2025)		✓	✓	ICLR'25
		HiddenDetectJiang et al. (2025)	✓	✓	✓	ACL'25
		SafePTRChen et al. (2025a)	✓	✓	✓	NeurIPS'25
		ALFARAn et al. (2025)	✓	✓	✓	NeurIPS'25
		SASAWang et al. (2025c)	✓	✓	✓	ACM MM'25
		CIDERXu et al. (2024b)		✓	✓	EMNLP'25 Findings
	Output-level Control	MLLM-ProtectorPi et al. (2024)	✓	✓		EMNLP'24
		ECSOGou et al. (2024)		✓		ECCV'24
		IMMUNEGhosal et al. (2025)	✓	✓	✓	CVPR'25
		DeGFZhang et al. (2025a)		✓		ICLR'25

Table 3: Recent advancements in multi-modal safety solutions. Cap. 1: Modality-Agnostic Readiness; Cap. 2: Cross-Modal Safety; Cap. 3: Internal Safety Intervention.

classic threat assumptions. (2) **Cross-Modal Safety:** Designing and ensembling modality-isolated safety mechanisms can disrupt cross-modal interactions and introduce unintended inconsistencies. Instead, safety solutions should operate holistically across modalities to preserve semantic coherence and support effective alignment and fusion. (3) **Internal Safety Intervention:** Traditional safety mechanisms typically focus on monitoring model inputs and outputs. However, in multi-modal systems, threats can emerge at intermediate stages –modality alignment and fusion. To address such risks, safety solutions should intervene at internal stages, rather than relying solely on surface-level observations.

4.2 SAFETY FINE-TUNING

Safety fine-tuning adapts pre-trained models with safety-oriented data without full retraining, including modality-wise fine-tuning within individual modalities or components (e.g., vision encoders or LLMs) and cross-modal fine-tuning that aligns interactions across modalities.

Several studies adopt modality-wise fine-tuning as a cost-effective safety strategy. On the vision side, works such as Schlarmann et al. (2024); Mao et al. (2023); Wang et al. (2024c); Shi et al. (2024) improve robustness by fine-tuning the visual encoder to resist adversarial perturbations while preserving feature representations. Yu et al. (2024a) further leverage text embeddings to guide attention and correct adversarial visual shifts. In contrast, other approaches enhance overall VLM safety by intervening in the LLM component. For example, Chakraborty et al. (2024) apply textual unlearning to prevent harmful context from propagating in the LLM’s latent space and redirect it toward safer representations.

Other works investigate cross-modal fine-tuning with safety-aligned datasets. VLGuard (Zong et al., 2024) and Robust-VLGuard (Wang et al., 2025b) provide safety-aligned datasets with both aligned and misaligned image-text pairs, improving MLLMs resilience against Gaussian noise and jailbreaks. UNIGUARD (Oh et al., 2025) models both unimodal and cross-modal harmful signals and enforces interaction-aware defenses across modalities. ProEAT (Lu et al., 2025) applies adversarial training to both visual projection layers and LLM to enhance cross-modal consistency and robustness. BYE (Rong et al., 2025) filters poisoned fine-tuning samples by detecting anomalous low-entropy cross-modal attention patterns. Xu et al. (2025) guide the projection of visual embeddings during alignment using retrieved textual evidence to mitigate toxic-image-induced risks.

Beyond attack scenarios, recent studies Guan et al. (2024); Li et al. (2025b) show that hallucination in MLLMs is also strongly influenced by instruction-tuning data. LRV-Instruction Liu et al. (2023a);

2024b) reduces instruction bias by incorporating both positive and negative instruction samples. PerturboLLaVA Chen et al. (2025b) introduces adversarial instruction-vision conflicts to discourage over-reliance on language priors. REVERIE Fei et al. (2024) further incorporates reasoning supervision by requiring rationales for correct and incorrect responses, promoting more robust cross-modal reasoning.

4.3 PREFERENCE-BASED OPTIMIZATION

Preference-based optimization improves safety alignment by optimizing over relative preferences between model outputs, rather than explicit labels or losses, using human or AI feedback to favor preferred responses over rejected ones.

Recent studies adopt preference-based optimization to align MLLMs under adversarial settings. SPA-VL (Zhang et al., 2025c) introduces a large-scale safety alignment dataset with preferred and rejected responses for supervised fine-tuning and subsequent RLHF/Direct Preference Optimization (DPO) training. GuardReasoner-VL (Liu et al., 2025c) adopts a reason-then-moderate paradigm with online RL to detect hidden harmful content. Liu et al. (2025a) combine classic adversarial training with DPO to update the visual encoder against adversarial image attacks, improving multi-modal safety alignment. Similarly, Weng et al. (2025) propose an adversary-aware DPO framework, in which the reference model is adversarially trained to provide reliable supervision, and a min-max objective is applied with perturbations in both image and latent spaces.

Beyond human feedback, several studies explore RL from AI feedback for hallucination mitigation. HA-DPO Zhao et al. (2023b) and Silkier Li et al. (2023b) apply DPO to favor accurate over hallucinated responses using AI-generated feedback or stronger teacher model preferences. Challenging standard DPO assumptions, POVid Li et al. (2025d) constructs preference data solely from hallucinated responses, treating ground-truth instructions as preferred outputs, while HalDetect Cai et al. (2025b) introduces fine-grained hallucination-aware rewards and FDPO to leverage human feedback at finer granularity.

4.4 TRAINING-FREE SOLUTIONS

Training-free solutions improve MLLM safety at inference time without updating model parameters, operating at the input level to block harmful or adversarial content, the internal level to suppress unsafe behaviors via intermediate representations, and the output level to verify or revise generations for safer responses.

At the input level, Guo et al. (2025) use an auxiliary LLM as a vision-free filter to screen and block harmful prompts before they are passed to the multi-modal model. AdaShield Wang et al. (2024f) prepends model inputs with input-aware defense prompts that automatically and adaptively safeguard MLLMs against jailbreak attacks. DiffPure-VLM Wang et al. (2025b) and BlueSuffix Zhao et al. (2025) mitigate visual adversarial perturbations using diffusion-based models. BlueSuffix further improves safety by rewriting textual prompts with an LLM-based purifier and prepending LLM-generated suffixes to the input.

For intervention in internal layers, CIDER Xu et al. (2024b) detects image-induced jailbreaks by identifying inputs with abnormally low cross-modal semantic distances in the embedding space. InferAligner Wang et al. (2024b) and ASTRA Wang et al. (2025a) suppress jailbreak-related hidden representations using steering vectors, with InferAligner deriving safety steering vectors from aligned LLMs and ASTRA extracting adversarial steering vectors via image-based attribution. SASA Wang et al. (2025c) and ETA Ding et al. (2025) adopt multi-stage safety mechanisms. SASA projects semantically rich representations from deeper fused layers to early layers to enhance risk perception, while ETA combines shallow alignment with deep sentence-level selection to enforce safety. SafePTR Chen et al. (2025a) and HiddenDetect Jiang et al. (2025) partition models into vulnerable and safety-relevant layers. SafePTR prunes harmful cross-modal tokens in vulnerable layers, whereas HiddenDetect detects unsafe prompts by measuring the alignment of activations with a refusal-aware embedding across safety layers.

For output-level control, Pi et al. (2024) assess response harmfulness using a harm detector and revise unsafe outputs with a detoxifier. IMMUNE Ghosal et al. (2025) guides token selection during decoding using a learned safety reward model to favor safer continuations. Gou et al. (2024) enable self-checking in MLLMs and regenerate safer outputs by rerouting visual inputs through an alignment-protected text-only pipeline.

Multi-modal hallucination can be mitigated by guided or self-correcting decoding. DeGF Zhang et al. (2025a) verifies and corrects model responses using generative feedback from text-to-image models, while MARINE Zhao et al. (2024a) guides decoding with grounded object representations and classifier-free guidance to emphasize visual evidence. At the attention level, EAH Li et al. (2025c) and VHR Zhang et al. (2024b) mitigate visual attention sink by strengthening image-token attention in shallow layers, while An et al. (2025) adaptively rebalances attention in multi-modal RAG based on retrieval scores and query-context relevance. To reduce over-reliance on language priors, Woodpecker Yin et al. (2024b) detects and corrects hallucinations using visual experts, and Liu (2025) encourage greater reliance on visual context.

5 FUTURE DIRECTIONS

Despite progress, MLLM safety remains challenged by expanded attack surfaces, cross-modal interactions, and new constraints, motivating future research.

5.1 RESILIENCE TO PARTIAL CORRUPTION

As discussed in Section 3, partial corruption can undermine system safety. Although MLLMs face expanded attack surfaces and modality inconsistencies, they also benefit from overlapping cross-modal information that can support robustness when some modalities are compromised. To avoid new points of failure, models should not over-rely on any subset of modalities. Future defenses should instead exploit cross-modal redundancy through mechanisms such as selective modality rejection, confidence-aware fusion, or adaptive weighting, enabling models to down-weight unreliable modalities while preserving performance and robustness.

5.2 TOWARD UNIFIED SAFETY FRAMEWORK

Many existing methods apply modality-specific defenses independently, but such ensemble-style strategies are neither scalable nor well aligned with multi-modal principles. Since not all modalities are compromised simultaneously, uniform defenses are often unnecessary and inefficient, while aggressive filtering can disrupt cross-modal coherence and degrade alignment and fusion. Future work should pursue unified, modality- and threat-agnostic safety frameworks, particularly by extending defenses to shared embedding and fusion stages and explicitly accounting for modality alignment through alignment-aware training or cross-modal consistency regularization.

5.3 HALLUCINATION BEYOND TEXT

Hallucination remains a fundamental risk in MLLMs, and current approaches lack principled and scalable solutions. Future work should improve data quality and calibration across modalities, strengthen cross-modal alignment through enhanced representations and consistency regularization, and pursue architectural advances that mitigate hallucination at its source. Progress also depends on theoretically grounded, human-aligned, and truly multimodal evaluation benchmarks beyond simplified or model-based assessments.

5.4 FROM MANUAL TO AGENT-DRIVEN SAFETY

Existing AI safety frameworks largely rely on manual efforts to identify vulnerabilities and design safeguards, an approach that is increasingly insufficient for modern multi-modal systems. As discussed in Sections 3 and 4, the threat surface of MLLMs has expanded into a high-dimensional and dynamic space spanning multiple modalities, cross-modal interactions, and diverse attack patterns, making systematic manual exploration infeasible. Moreover, many defenses are evaluated under static threat assumptions or limited attack settings without considering adaptive adversaries (Nasr et al., 2025). To address these challenges, we advocate a shift toward automated, agent-driven red-blue teaming, where red-teaming agents autonomously explore complex attack strategies and discovered vulnerabilities guide blue-teaming mechanisms for the continuous development of more robust and proactive safety solutions.

6 CONCLUSION

In response to the evolving safety landscape, this survey introduces a new taxonomy of multi-modal safety risks – compromised modality integration, modality misalignment, and fused safety risks – and highlights shifts in threat assumptions under compositional architectures. We further summarize emerging safety constraints and recent advances in safety solutions, including safety fine-tuning, preference-based optimization, and training-free approaches, and outline future directions toward unified and robust safety frameworks for next-generation multi-modal AI systems.

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